

[64]



Seat No.: _____

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SARDAR PATEL UNIVERSITY

M.Sc (Information Technology) (Semester 1) April - 2023

PS01CINT53: Computer Networks & Cyber Security

Date: 17/04/2023 (Monday)

Time: 10:00 AM to 01:00 PM

Total: 70 Marks

Q-1 Multiple Choice Questions.

[08]

- 1 _____ of OSI model is responsible for row bit transmission.
(A) session layer (B) physical layer
(C) data link layer (D) presentation layer
- 2 _____ consists of two distinct components: transmission lines & switching elements.
(A) Host (B) Subnet
(C) Server (D) Intermediate server
- 3 _____ routing algorithms do not base their routing decisions on measurements or estimates of the current traffic and topology.
(A) Dynamic (B) Static
(C) Flooding (D) Shortest Path
- 4 The _____ satellites are so close to the earth, the ground stations do not need much power, and the round-trip delay is only a few milliseconds.
(A) LEO (B) MEO
(C) GEO (D) None of these
- 5 In a _____ handoff, the telephone is acquired by the new base station before the previous one signs off.
(A) soft (B) Hard
(C) Mutual (D) None
- 6 A network Built up from company computer and leased telephone lines is called _____.
(A) Public Network (B) Private network
(C) Both (D) None
- 7 Substitution ciphers preserve the order of the plaintext symbols but disguise them.
(A) Substitution (B) Transposition
(C) Monoalphabetic (D) None
- 8 The art of breaking ciphers is called _____.
(A) Cipher text (B) Plain Text
(C) Crypt analysis (D) None

- Q-2 Answer the Following short questions. (Attempt Any SEVEN)** [14]
1. List Advantages and disadvantages of Fiber.
 2. How can we classify Computer Network?
 3. Give any three differences between Fiber and satellite.
 4. Draw a diagram of Multistation Gigabit Ethernet configuration.
 5. Full form of GSM & CDMA
 6. What is Tunneling?
 7. Define Cryptography & Firewall
 8. List the types of cybercrimes.
 9. Define plaintext and cipher text.
- Q-3 Answer the following Questions.**
- (A) Explain OSI model in detail. 6
- (B) List out network devices and explain Repeater and Router. 6
- OR
- (B) Explain Twisted pair cable in detail. 6
- Q-4 Answer the following Questions.**
- (A) Explain Geosynchronous satellite and Medium Earth Orbit satellites. 6
- (B) Explain Flooding Routing Algorithm. 6
- OR
- (B) Explain Shortest path Routing Algorithm. 6
- Q-5 Answer the following Questions.**
- (A) What is Fragmentation? Explain types of Fragmentation in detail. 6
- (B) Write a short note on Advanced Mobile Phone System. 6
- OR
- (B) Explain Virtual private network in details. 6
- Q-6 Answer the following Questions.**
- (A) What is firewall? Explain Packet filter firewall in detail. 6
- (B) What is Symmetric key algorithms in detail. 6
- OR
- (B) Define cryptography. Explain Substitution Ciphers in details 6

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